

EE3621: Digital Signal Processing

Lecture 9: Fast Fourier Transform (FFT) — Decimation in Time (DIT) (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Welcome, Unit II Intro. The problem with Direct DFT — Computational Complexity Motivation.
- **05:00 – 15:00 (10 mins):** The core idea of FFT: Divide and Conquer strategy. Decimating the time-domain signal into even and odd indices.
- **15:00 – 20:00 (5 mins):** Twiddle factor mathematics and reduction proofs.
- **20:00 – 25:00 (5 mins):** The 2-point Butterfly operation. Understanding the basic building block.
- **25:00 – 30:00 (5 mins):** Signal Flow Graph for $N = 8$, Bit-Reversal permutation, and intuitive understanding.
- **30:00 – 35:00 (5 mins):** Complexity Analysis and Comparison Table. Implementing IFFT via FFT.
- **35:00 – 40:00 (5 mins):** Checkpoint questions with detailed answers and conclusion.

2 Computational Complexity Motivation

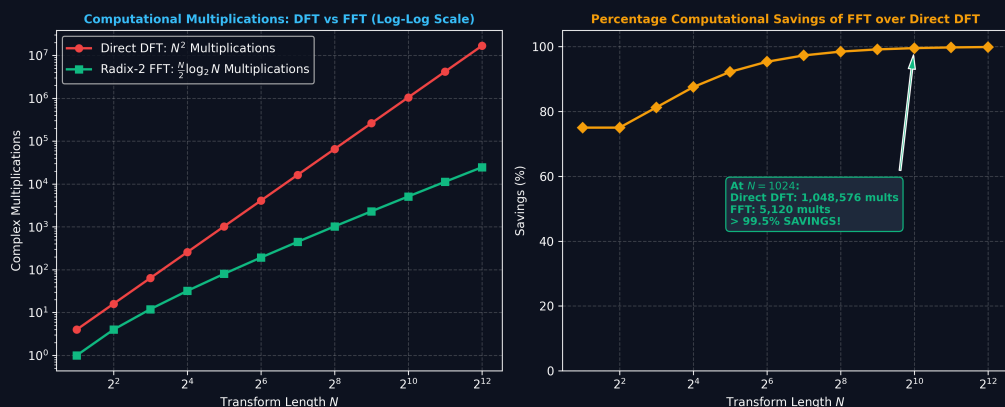


Figure 1: Computational Scaling: Direct DFT $O(N^2)$ vs. Radix-2 FFT $O(N \log_2 N)$ (> 99.5% Savings for $N = 1024$).

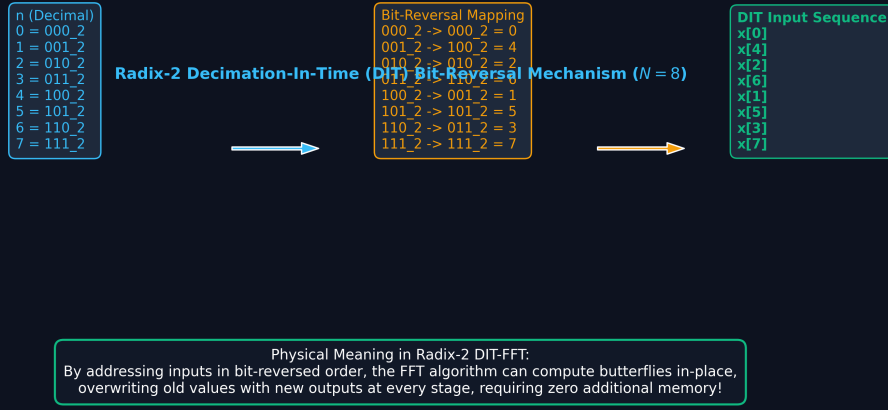


Figure 2: Radix-2 Decimation-in-Time Bit-Reversal Address Mapping for In-Place Memory Execution.

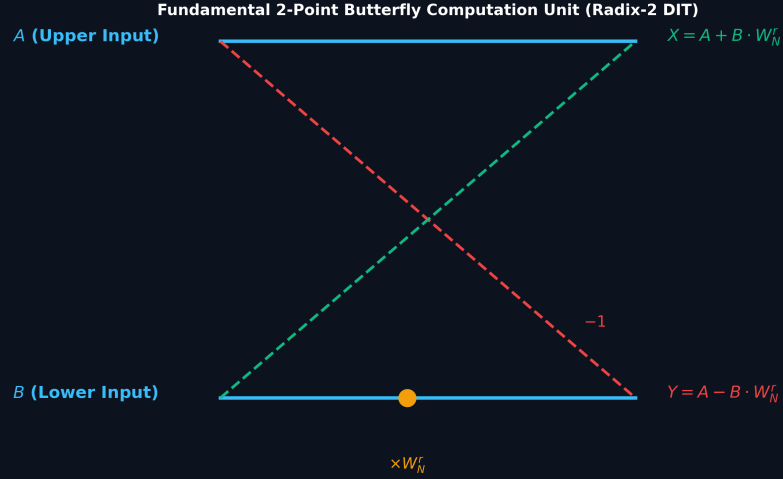


Figure 3: Fundamental 2-Point Butterfly Computation Unit (Radix-2 DIT).

The Direct Discrete Fourier Transform (DFT) of an N -point sequence $x[n]$ is given by:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad (1)$$

where $W_N = e^{-j\frac{2\pi}{N}}$ is the twiddle factor, and $0 \leq k \leq N - 1$.

Let's carefully count the operations for the direct calculation:

1. For each frequency bin $X[k]$, we must multiply N complex numbers $x[n]$ by N complex twiddle factors W_N^{kn} . This requires N complex multiplications.
2. Summing these N terms requires $N - 1$ complex additions.
3. Since there are N frequency bins (from $k = 0$ to $N - 1$), the total operations are:

- **Total Complex Multiplications:** $N \times N = N^2$
- **Total Complex Additions:** $N \times (N - 1) = N^2 - N$

For small N , this is fine. But consider a typical audio block of $N = 1024$: Number of complex multiplications $= 1024^2 = 1,048,576 > 10^6$. Since one complex multiplication requires 4 real multiplications and 2 real additions, the burden is massive. Real-time processing becomes impossible for longer sequences without a better algorithm.

The Fast Fourier Transform (FFT) reduces this complexity from $O(N^2)$ to $O(N \log_2 N)$. For $N = 1024$, the FFT requires only $\frac{N}{2} \log_2 N = 512 \times 10 = 5120$ multiplications.

3 Divide & Conquer Decomposition: Decimation in Time (DIT)

The Radix-2 DIT FFT algorithm works by recursively dividing an N -point sequence (where N is a power of 2) into two shorter sequences of length $N/2$.

Let us separate $x[n]$ into its even-indexed and odd-indexed samples:

- Even samples: $x_e[m] = x[2m]$ for $m = 0, 1, \dots, \frac{N}{2} - 1$
- Odd samples: $x_o[m] = x[2m + 1]$ for $m = 0, 1, \dots, \frac{N}{2} - 1$

Substitute this into the DFT equation:

$$X[k] = \sum_{m=0}^{N/2-1} x[2m] W_N^{2mk} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{(2m+1)k} \quad (2)$$

Factor out W_N^k from the second summation:

$$X[k] = \sum_{m=0}^{N/2-1} x[2m] W_N^{2mk} + W_N^k \sum_{m=0}^{N/2-1} x[2m+1] W_N^{2mk} \quad (3)$$

This is the foundational equation of the DIT FFT!

4 Twiddle Factor Reduction

Theorem: $W_N^2 = W_{N/2}$

Proof:

$$W_N^2 = \left(e^{-j \frac{2\pi}{N}} \right)^2$$

$$W_N^2 = e^{-j \frac{4\pi}{N}}$$

$$W_N^2 = e^{-j \frac{2\pi}{N/2}}$$

$$W_N^2 = W_{N/2}$$

Applying this identity: $W_N^{2mk} = (W_N^2)^{mk} = W_{N/2}^{mk}$, we rewrite our split DFT equation:

$$X[k] = \sum_{m=0}^{N/2-1} x_e[m] W_{N/2}^{mk} + W_N^k \sum_{m=0}^{N/2-1} x_o[m] W_{N/2}^{mk} \quad (4)$$

The first sum is the $N/2$ -point DFT of the even sequence, $G[k]$. The second sum is the $N/2$ -point DFT of the odd sequence, $H[k]$.

$$\textbf{KEY RESULT: } X[k] = G[k] + W_N^k H[k] \quad (5)$$

For the upper half of the spectrum ($k \geq N/2$), we use the symmetry property $W_N^{k+N/2} = -W_N^k$:

$$X[k + N/2] = G[k] - W_N^k H[k] \quad (6)$$

5 The Butterfly Operation

The equations:

$$X[k] = G[k] + W_N^k H[k] \quad (7)$$

$$X[k + N/2] = G[k] - W_N^k H[k] \quad (8)$$

form the **Butterfly Operation**. If we let $A = G[k]$ and $B = H[k]$, the outputs A' and B' are:

$$A' = A + W_N^k B \quad (9)$$

$$B' = A - W_N^k B \quad (10)$$

Engineering Intuition: Instead of multiplying B by W_N^k and separately by $-W_N^k$, we calculate $(W_N^k B)$ once, add it to A , and subtract it from A . This halves multiplications.

6 Signal Flow Graph & Bit-Reversal Permutation

For $N = 8$, the sequence is recursively divided until 2-point DFTs are reached. The final input order is: $x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]$.

Why this order? Observe the binary index:

- $1 \rightarrow 001 \rightarrow \text{reversed: } 100 \rightarrow 4$
- $2 \rightarrow 010 \rightarrow \text{reversed: } 010 \rightarrow 2$
- $3 \rightarrow 011 \rightarrow \text{reversed: } 110 \rightarrow 6$

This is the **Bit-Reversal Permutation**.

6.1 The 8-Point DIT-FFT Stages

- **Stage 1:** Four 2-point butterflies.
- **Stage 2:** Two 4-point FFT blocks using W_8^0 and W_8^2 .
- **Stage 3:** One 8-point FFT block using $W_8^0, W_8^1, W_8^2, W_8^3$.

7 Complexity Analysis

Let $T(N)$ be the number of complex multiplications for an N -point FFT. An N -point DFT requires:

1. Two $N/2$ -point DFTs: $2T(N/2)$
2. $N/2$ multiplications by W_N^k for the butterflies.

Recurrence relation:

$$T(N) = 2T(N/2) + \frac{N}{2} \quad (11)$$

Proof by Recursion:

$$T(N) = 2 \left(2T(N/4) + \frac{N}{4} \right) + \frac{N}{2} = 4T(N/4) + 2\frac{N}{2}$$

After $\nu = \log_2 N$ stages, we reach $T(1) = 0$:

$$T(N) = 2^\nu T(N/2^\nu) + \nu \frac{N}{2} = \frac{N}{2} \log_2 N \quad (12)$$

8 Comparison Table: DFT vs FFT

Signal Length N	Direct Multiplications	FFT Multiplications	Speedup Factor
8	64	12	5.3x
64	4,096	192	21.3x
256	65,536	1,024	64.0x
1024	1,048,576	5,120	204.8x
4096	16,777,216	24,576	682.7x

Table 1: Complexity comparison

9 Computing IFFT using FFT

The Inverse DFT (IDFT) is:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \quad (13)$$

Using the **conjugate trick**, take the complex conjugate of $x[n]$:

$$x^*[n] = \frac{1}{N} \sum_{k=0}^{N-1} X^*[k] W_N^{kn} = \frac{1}{N} \text{FFT}\{X^*[k]\} \quad (14)$$

Taking the conjugate again:

$$x[n] = \frac{1}{N} (\text{FFT}\{X^*[k]\})^* \quad (15)$$

Physical Intuition: Conjugate the frequency data, run forward FFT, conjugate output, and divide by N .

10 Checkpoint Questions

Q1: In an 8-point Radix-2 DIT FFT, what is the index for the 5th element?

Answer: Normal index 4 $\rightarrow$ binary 100₂ $\rightarrow$ reverse 001₂ $\rightarrow$ decimal 1. It is $x[1]$.

Q2: At 20 ns per complex multiplication, calculate time saved by FFT for $N = 4096$.

Answer:

- Direct: $4096^2 \times 20 \text{ ns} = 335.54 \text{ ms}$
- FFT: $24576 \times 20 \text{ ns} = 0.491 \text{ ms}$
- Saved: $\sim 335.05 \text{ ms}$.

Q3: Prove $W_N^{k+N/2} = -W_N^k$ and its use.

Answer: $W_N^{k+N/2} = W_N^k \cdot W_N^{N/2} = W_N^k \cdot e^{-j\pi} = -W_N^k$. Used in the butterfly operation to reduce multiplications by sharing terms.

11 Table of Key Formulas

Concept	Formula / Property
Direct DFT	$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$
Twiddle Factor	$W_N = e^{-j\frac{2\pi}{N}}$
Symmetry Property	$W_N^{k+N/2} = -W_N^k$
Periodicity Property	$W_N^{k+N} = W_N^k$
Square Property	$W_N^2 = W_{N/2}$
DIT Decomposition	$X[k] = G[k] + W_N^k H[k]$
FFT Multiplications	$\frac{N}{2} \log_2 N$
IFFT via FFT	$x[n] = \frac{1}{N} (\text{FFT}\{X^*[k]\})^*$